

Are Urban Demand Zones Stable? Distributed Spatial Clustering of 142 Million NYC Taxi Trips on Google Cloud

PSTAT 135 Final Project — Shahil (shahil@ucsb.edu) Dataset: `bigquery-public-data.new_york.tlc_yellow_trips_2015` · Stack: BigQuery · Dataproc/PySpark · BigQuery ML · NetworkX

Abstract

Vehicle-routing problems (VRP) are NP-hard, but a common tractable heuristic decomposes a city into demand *zones* and routes within each. A natural hypothesis in operations research is that these zones should be **time-adaptive** — re-optimized for each operational period — because demand geography shifts between morning rush, midday, evening rush, and night. We test this hypothesis empirically at scale: 142,567,774 cleaned 2015 NYC yellow-taxi trips, clustered with distributed K-means (Apache Spark on Google Cloud Dataproc) into 20 zones for each of four time periods, with a full-population routing simulation comparing static vs. time-adaptive zoning. **Contrary to the hypothesis, we find that optimal zones are remarkably stable:** centroids shift only 0.5–1.0 km between periods, and time-adaptive zoning reduces mean dispatch distance by only ~1% (range -1.4% to +2.8%) versus a fixed design. The cause is demand concentration — NYC yellow-cab pickups are so dominated by the Manhattan core that 20 zones cover them well in every period. We complement this with PageRank hub analysis (validating Midtown as the network center) and BigQuery ML demand forecasting (a regularized linear model reaches $R^2=0.06$, a boosted tree $R^2=0.87$). The result is a cautionary, empirically-grounded counterpoint to the assumption that temporal zone adaptation yields large routing gains. An interactive Cloud-hosted explorer — rank-ordered demand zones, an animated 24-hour demand heatmap, and a zone-drift overlay — accompanies the analysis (§4.7). A longitudinal five-arm controlled extension (§4.8) comparing Yellow 2015, Green 2015, Yellow 2024, and FHV 2019/2024 on the same 263 TLC zones reveals that the relationship between demand dispersion and adaptive-zoning value is **non-monotone**: static zoning is optimal at both extremes (concentrated Yellow '15 and fully dispersed FHV '24), while time-volatile intermediate regimes — outer-borough taxis and today's reduced yellow-cab fleet — gain most from adaptation.

1. Introduction

Urban logistics and ride-dispatch operations rest on the vehicle-routing problem, which is NP-hard in general. A standard engineering response is **zone decomposition**: partition the service area into a small number of zones, assign vehicles to zones, and solve a far smaller routing problem within each. The quality of this decomposition hinges on where the zones are placed.

It is widely assumed that good zones must be **time-adaptive** — that the partition optimal for the 8 a.m. Financial District commute is wasteful at 2 a.m. when demand migrates to nightlife districts. If true, this motivates time-varying VRP decomposition and dynamic dispatch. **This project asks whether that assumption actually holds for a real, large-scale demand dataset**, and quantifies the routing efficiency at stake.

Our contribution is threefold: 1. A reproducible, fully cloud-native pipeline that clusters 142M trips with distributed K-means — an analysis infeasible on a single machine. 2. A fair, full-population routing simulation (fixed number of zones) measuring the dispatch-distance gain of adaptive vs. static zoning. 3. An honest, somewhat counterintuitive finding: **zones are stable and adaptivity buys almost nothing** for this dataset, with a clear structural explanation.

2. Data

Source. `bigquery-public-data.new_york.tlc_yellow_trips_2015`, the NYC Taxi & Limousine Commission trip records. 2015 is the last full year of peak yellow-cab dominance before ride-hailing structurally reduced demand, and the highest-volume year with raw pickup/dropoff latitude-longitude (later years publish only zone IDs, which would preclude true spatial clustering).

Cleaning. We retained trips within an NYC bounding box (lat 40.5–40.9, lon -74.3 to -73.7 for both pickup and dropoff), fares $\backslash \$2.50$ – $\backslash \$100$, and trip distance 0.1–50 mi, yielding **142,567,774 trips** (Jan 1 – Dec 31 2015; 0 null coordinates; mean 2.97 mi, $\backslash \$12.79$). The cleaned table is 11 columns.

Derived tables (BigQuery SQL). `zone_demand` (563,696 rows — hourly demand per ~1 km grid cell), `trips_with_period` (142.5M rows, adds `time_period` and `day_type` labels), `od_matrix` (45,477 origin→destination edges with ≥ 20 trips).

Time periods. AM_PEAK (7–9 a.m.), MIDDAY (10 a.m.–3 p.m.), PM_PEAK (4–7 p.m.), NIGHT (8 p.m.–6 a.m.). Distribution by volume (Fig. 0): NIGHT is largest (52.3M) with the longest mean trip (3.30 mi — late-night airport/outer-borough runs), AM_PEAK smallest (18.3M).

Architecture. Public BigQuery → cleaned `taxi_analysis` dataset → Parquet export to Cloud Storage (293 shards, 2.16 GiB) → Dataproc/PySpark clustering → results back to GCS/BigQuery → static figures plus an interactive Google Maps explorer hosted on GCS (rank-ordered demand bubbles, an animated 24-hour demand-grid heatmap, and a zone-drift overlay; see §4.7).

3. Methodology

Time-stratified K-means (Dataproc/PySpark). For each period we standardize (`pickup_lat`, `pickup_lon`), then fit K-means. Model selection used a silhouette + elbow sweep over $k \in \{10, 15, 20, 25\}$ per period (Fig. 1); silhouette scores were stable (~ 0.51 – 0.60), so for the cross-period comparison we **fix $k = 20$ for every period**. This fixed- k choice is essential: with different k per period, the period with more centroids trivially achieves shorter nearest-centroid distance, confounding any static-vs-adaptive comparison. Clustering ran on a 6-worker `n1-standard-4` cluster.

Zone-stability metric. For each pair of periods we compute an average (symmetric) nearest-centroid displacement in kilometers — a Hausdorff-inspired drift measure (Fig. 3): for centroid sets C_1, C_2 , $\frac{1}{2} \cdot (\text{mean over } C_1 \text{ of min haversine to } C_2 + \text{mean over } C_2 \text{ of min haversine to } C_1)$.

Routing simulation (the key OR result). For every one of the 142M trips we compute the haversine distance to its nearest zone centroid under (a) **adaptive** zoning — that period's own $k=20$ centroids — and (b) **static** zoning — a fixed baseline period's (AM_PEAK) centroids used all day. This was implemented as a vectorized Spark `pandas_udf` broadcasting the ≤ 20 centroids and computing pairwise haversine inside each Arrow batch, so it scales to the full population rather than a small sample. We report the mean per period and the percentage reduction.

Graph analytics. From `od_matrix` we build a weighted directed graph (831 nodes, 31,480 edges with ≥ 50 trips) and compute PageRank ($\alpha=0.85$) to identify high-centrality hubs (Fig. 5).

Demand forecasting (BigQuery ML). We predict hourly `trip_count` per grid cell with three models: a regularized linear regression ($L1=L2=0.1$, the Lecture-7 penalized-regression connection), the same with temporal features one-hot encoded, and a boosted-tree regressor (Fig. 6).

4. Results

4.1 Exploratory analysis

Demand follows the expected bimodal weekday rhythm with AM and PM peaks, a Friday/Saturday weekly maximum, and mild seasonal variation across 2015 (Fig. 0). Pickups are overwhelmingly concentrated in Manhattan in every period — already foreshadowing the stability result.

4.2 Temporal clustering

Per-period K-means produces sensible 20-zone partitions (Fig. 2). Overlaying all four periods' centroids (Fig. 2b) shows them occupying nearly the same Manhattan core — a direct visual of stability.

4.3 Zone stability (drift)

Centroids move only **0.5–1.0 km** between periods (Fig. 3). The largest drift is **MIDDAY ↔ NIGHT = 1.02 km** (midday commerce vs. night nightlife), the smallest PM_PEAK ↔ MIDDAY = 0.52 km. Zones *do* shift, but modestly.

4.4 Key OR result — static vs. adaptive routing

On the full 142M-trip population, with a fixed 20 zones, time-adaptive zoning barely beats static (Fig. 4):

Period	adaptive (km)	static / AM zones (km)	gain
AM_PEAK	0.576	0.576	0.0% (baseline)
PM_PEAK	0.556	0.571	+2.8%
MIDDAY	0.579	0.571	-1.4%
NIGHT	0.616	0.625	+1.4%

Average gain $\approx$ 0.9%. One period is even slightly negative (within the noise of K-means minimizing squared distance in standardized space versus our mean-haversine metric). **The hypothesized large routing benefit of time-adaptive zoning does not materialize.** The structural reason is demand concentration: with pickups so dominated by the Manhattan core, any reasonable placement of 20 zones covers demand well, so the ~1 km of genuine zone drift is too small to change which zone is nearest for most trips.

4.5 Graph analytics

PageRank ranks the trip network's top hubs entirely within Midtown Manhattan — (40.76, -73.98) Times Square/Midtown, (40.75, -73.99) Penn Station/Herald Square (Fig. 5). This matches ground truth and validates the OD-graph construction.

4.6 Demand forecasting

A regularized **linear** model explains only $R^2=0.055$ ($MAE \approx 390$ trips); one-hot encoding the temporal features barely helps ($R^2=0.059$). A **boosted tree** reaches $R^2=0.871$ ($MAE \approx 92$) (Fig. 6). The gap is itself the lesson: demand is a strongly *nonlinear* function of location (the tree splits on lat/lon) and exhibits the bimodal hourly pattern a linear term cannot capture. Tree feature importance is shared roughly equally across longitude, day-of-week, hour, and latitude.

4.7 Interactive explorer

A browser-based companion (Google Maps JavaScript API, hosted on Cloud Storage) lets a reader interrogate the result directly through three linked views. **Bubbles** renders each period's 20 zones as rank-ordered markers, sized and coloured by pickup volume (drawn at constant on-screen size so they stay legible at any zoom). **Heatmap** shows a continuous ~ 1 km demand surface — a self-contained canvas overlay (Google removed the Maps `HeatmapLayer` in API v3.65) driven by an hourly demand grid aggregated from `zone_demand` into 24 frames; a slider and play button animate the city's full 24-hour demand pulse, contracting to the Manhattan core before dawn (hour 05, the daily minimum) and blooming outward at the evening peak (hour 19, 8.8M pickups). **Overlay** traces each AM_PEAK zone to its nearest centroid in the other three periods and draws the resulting drift path, coloured green (stable) $\rightarrow$ red (mover): almost every path is a short green stub — an interactive confirmation of the central finding (mean traced drift 0.66 km, max 1.66 km) — while the few genuine movers reach toward the airports. The explorer thus makes the stability result tangible rather than merely tabulated. Deep-links (`#heatmap`, `#overlay`, `#h=19`) jump straight to a view or hour.

4.8 The rideshare era: a controlled natural experiment

The 2015 finding rests on demand *concentration*. But 2015 yellow taxis were Manhattan-bound by regulatory design: the NYC Taxi & Limousine Commission restricted yellow-cab street hails to Manhattan and the airports, so comparing raw 2015 yellow pickups to modern Uber/Lyft pickups would confound *mode/coverage* with *time*. A simple two-arm comparison cannot disentangle whether demand became more dispersed because of rideshare *or* simply because the two services have different legal footprints. We address this with a **five-arm controlled design** that separates the two effects by keeping all arms on the same 263 TLC taxi zones and fixing $k=20$ throughout.

Study arms.

Arm	Mode	Year	Role
A Yellow '15	Yellow taxi	2015	Pre-rideshare baseline (142.5M trips)

Arm	Mode	Year	Role
B Green '15	Green boro-taxi	2015	Outer-borough taxi control (19.2M trips)
C Yellow '24	Yellow taxi	2024	Mode control — same service, 9 yrs later (41.0M trips)
D FHV '19	Uber/Lyft/Via	2019	Early rideshare trajectory (44.1M trips, Feb–Mar)
E FHV '24	Uber/Lyft/Via	2024	Modern rideshare reality (240.1M trips)

A→C isolates *time* within one mode; C→E isolates *coverage/mode* within one year. Every arm is additionally re-run restricted to Manhattan-only zones (the common footprint all services operate in) as a further coverage control. Note: arm D covers only February–March 2019 due to mid-year schema changes in the TLC parquet files; it is treated as an early-rideshare snapshot rather than a full-year estimate.

The dispersion shift is real (Fig. 7). Manhattan's pickup share fell from 92.4% (Yellow '15) to 38.8% (FHV '24), with Brooklyn and the Bronx absorbing the overflow. But crucially, Yellow '24 remains nearly as Manhattan-concentrated as Yellow '15 (88.7%), confirming the geographic shift is driven by rideshare's city-wide footprint, not by a general change in travel demand.

Results (Fig. 8–9).

Arm	Manhattan %	Radius km	Drift km	Adaptive gain	(Manhattan-only)
Yellow '15	92.4%	3.10	0.70	+0.18%	+2.11%
Green '15	28.4%	6.64	1.08	+7.50%	−8.75%
Yellow '24	88.7%	3.89	1.28	+9.12%	+12.70%
FHV '19	44.0%	7.83	1.67	+4.29%	+2.77%
FHV '24	38.8%	8.39	1.48	−1.53%	+2.06%

The finding is more nuanced than the initial hypothesis (Fig. 10). The naive prediction — "more dispersion → more adaptive gain" — does not hold monotonically. Three observations stand out:

- 1. Time-within-yellow (A→C):** Yellow '24 remains nearly as Manhattan-concentrated as Yellow '15 (88.7% vs 92.4%), yet adaptive gain jumps from +0.2% to +9.1%. Something other than total dispersion changed — post-pandemic commute patterns altered the intra-day geography of yellow-cab demand even though the aggregate footprint did not.
- 2. Mode/coverage (C→E, same year):** Rideshare FHV '24 is far more dispersed than Yellow '24 but has *lower* adaptive gain (−1.5% vs +9.1%). When demand is so evenly spread across all 263 zones that no set of 20 centroids is clearly "wrong," static zoning is already

near-optimal and adaptation buys nothing. The Manhattan-only control confirms a small residual gain (+2.1%) exists within the common footprint, but it is modest.

3. **Green boro-taxi (B):** Outer-borough taxis show the largest all-NYC adaptive gain (+7.5%) because outer-borough demand shifts dramatically between periods (nightlife, airports, shift workers) — the type of volatile, time-varying demand where adaptive zoning pays off most.

The practical decision rule (Fig. 10) is therefore not simply "adapt when dispersed" but **"adapt when demand shifts substantially by time period"** — concentrated demand that is temporally stable (Yellow '15) and fully dispersed uniform demand (FHV '24) both favour static zoning; it is the intermediate, time-volatile regime that favours adaptation. The rideshare era partially entered this regime (FHV '19: +4.3%) but by 2024 demand is so city-wide and uniform that static zoning is again robust — and a smaller fleet of yellow cabs now shows the largest time-of-day sensitivity of all.

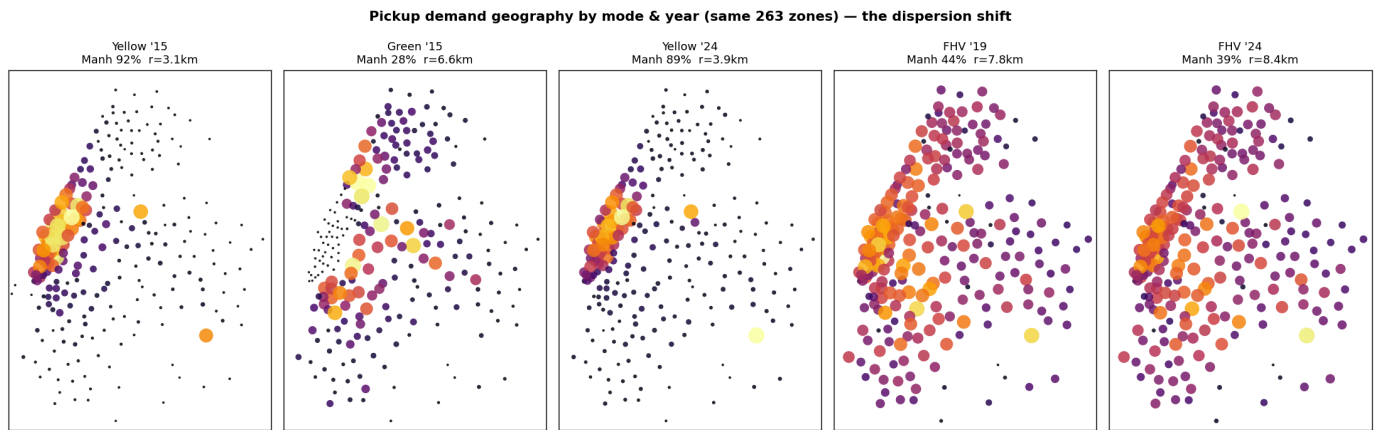


Fig. 7 — Pickup demand geography across all five arms (same 263 zones). Yellow '24 remains Manhattan-bound; rideshare spread city-wide by 2024.

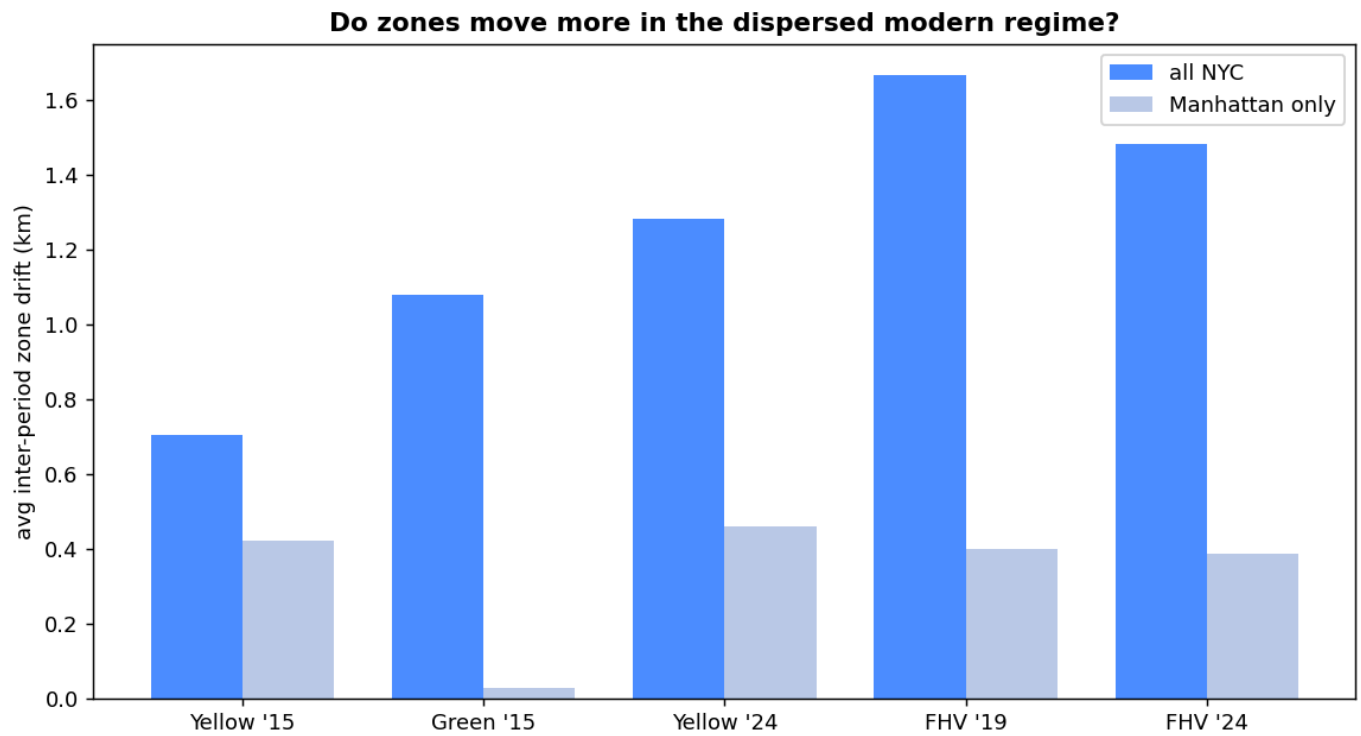


Fig. 8 — Average inter-period zone drift by arm (all NYC and Manhattan-only). Modern yellow taxi and early FHV drift most between time periods.

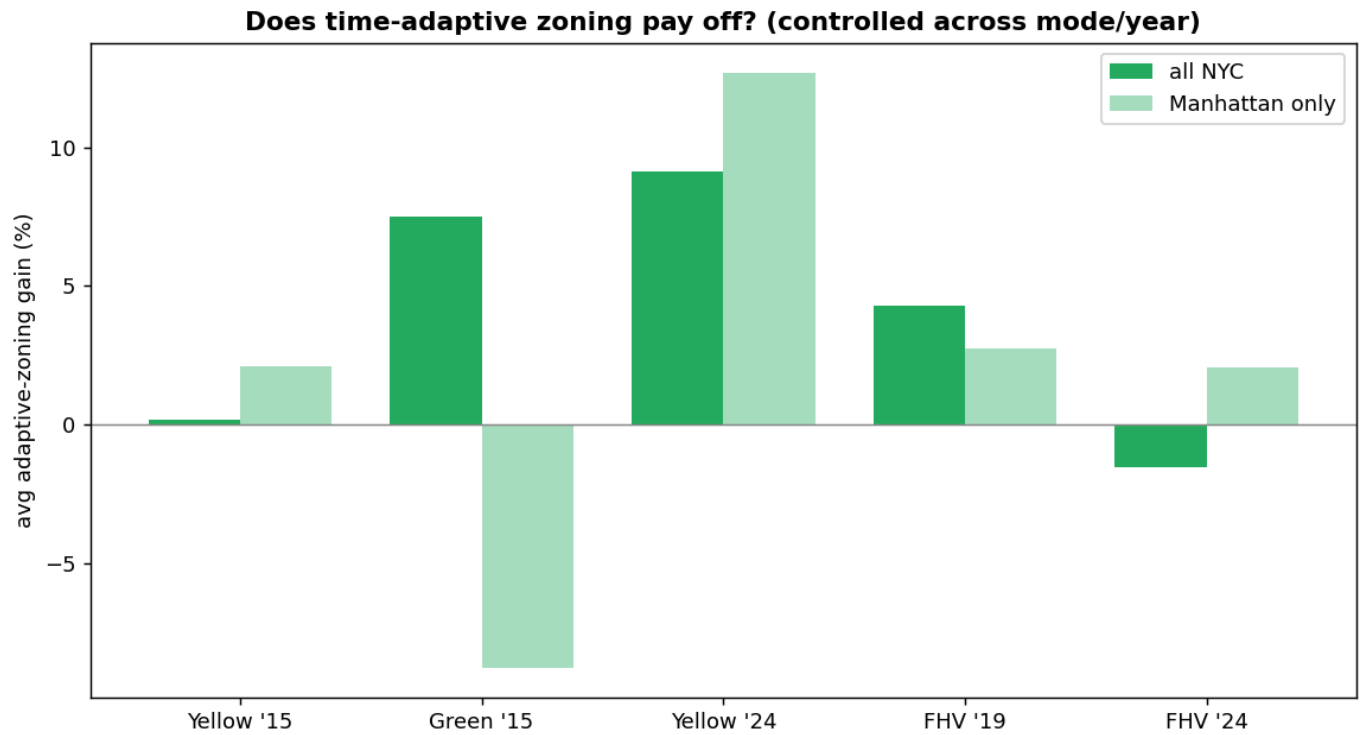
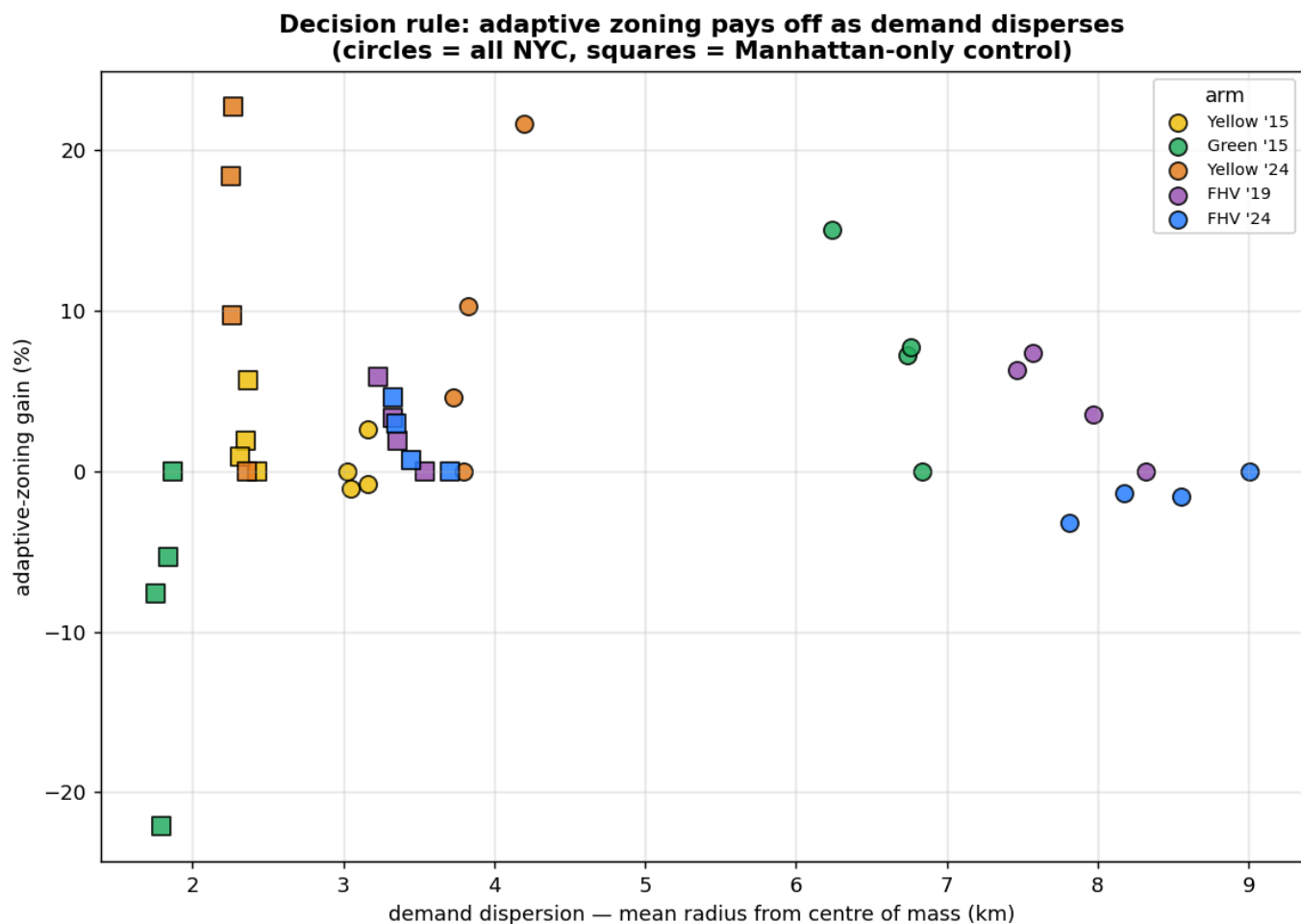


Fig. 9 — Static-vs-adaptive dispatch-distance gain by arm. Yellow '24 shows the largest gain despite being barely more dispersed than Yellow '15.



shifts more by time than the Manhattan core — may reveal where adaptivity does pay off; (c) a cross-year comparison (2014–2017) could track how the ride-hailing disruption reshaped these zones; (d) real-time adaptive zoning via streaming (Cloud Pub/Sub) would extend the dispatch application.

References

[1] Bramel, J. & Simchi-Levi, D. (1997). *The Logic of Logistics*. Springer. [2] Zaharia, M. et al. (2016). Apache Spark: A Unified Engine for Big Data Processing. *CACM* 59(11). [3] NYC TLC Trip Record Data. <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page> [4] Page, L. et al. (1999). The PageRank Citation Ranking. Stanford TR. [5] Ester, M. et al. (1996). DBSCAN. *KDD-96*. [6] Tibshirani, R. (1996). Regression Shrinkage and Selection via the Lasso. *JRSS-B* 58(1). [7] Agatz, N. et al. (2012). Optimization for Dynamic Ride-Sharing. *Transportation Science* 46(3). [8] Google Cloud BigQuery ML Docs. <https://cloud.google.com/bigquery/docs/bqml-introduction> [9] Matter, U. (2024). *Big Data Analytics*. CRC Press. [10] Rioux, J. (2022). *Data Analysis with Python and PySpark*. O'Reilly.

Appendix — Figures

- **Fig. 0** `fig0_eda_overview.png` — hourly/DOW/monthly volume + pickup scatter by period
- **Fig. 1** `fig1_elbow_silhouette.png` — elbow (WSSSE) + silhouette vs k, per period
- **Fig. 2** `fig2_cluster_scatter_4panel.png` — 20 zones per period; **Fig. 2b** `fig2b_centroid_overlay.png` — all overlaid
- **Fig. 3** `fig3_zone_drift.png` — centroid drift heatmap
- **Fig. 4** `fig4_static_vs_adaptive.png` — routing: adaptive $\approx$ static
- **Fig. 5** `fig5_pagerank.png` — top OD hubs (Midtown)
- **Fig. 6** `fig6_demand_models.png` — BQML model comparison + tree feature importance
- **Fig. 7** `figA_era_demand.png` — 5-arm demand geography (dispersion shift)
- **Fig. 8** `figB_era_drift.png` — 5-arm zone-drift comparison
- **Fig. 9** `figC_era_gain.png` — 5-arm static-vs-adaptive gain
- **Fig. 10** `figD_gain_vs_dispersion.png` — decision rule: gain vs dispersion (non-monotone)
- **Interactive explorer:** <https://storage.googleapis.com/pstat135-taxi-shahil/viz/index.html> — three linked views: rank-ordered demand *bubbles*, an animated 24-hour demand-grid *heatmap*, and a zone-drift overlay (deep-links `#heatmap`, `#overlay`, `#h=19`).

Code: `o3_eda.py · o4_temporal_kmeans.py · o4c_cluster_scatter.py · fig1_silhouette.py · o5_routing_plot.py · o6_zone_drift.py · o7_pagerank.py · o8_bqml_demand.sql ·`

viz/index.html (with viz/heat_hourly.json — the 24-hour ~1 km demand grid aggregated from zone_demand). Full numbers in RESULTS.md.

Figures

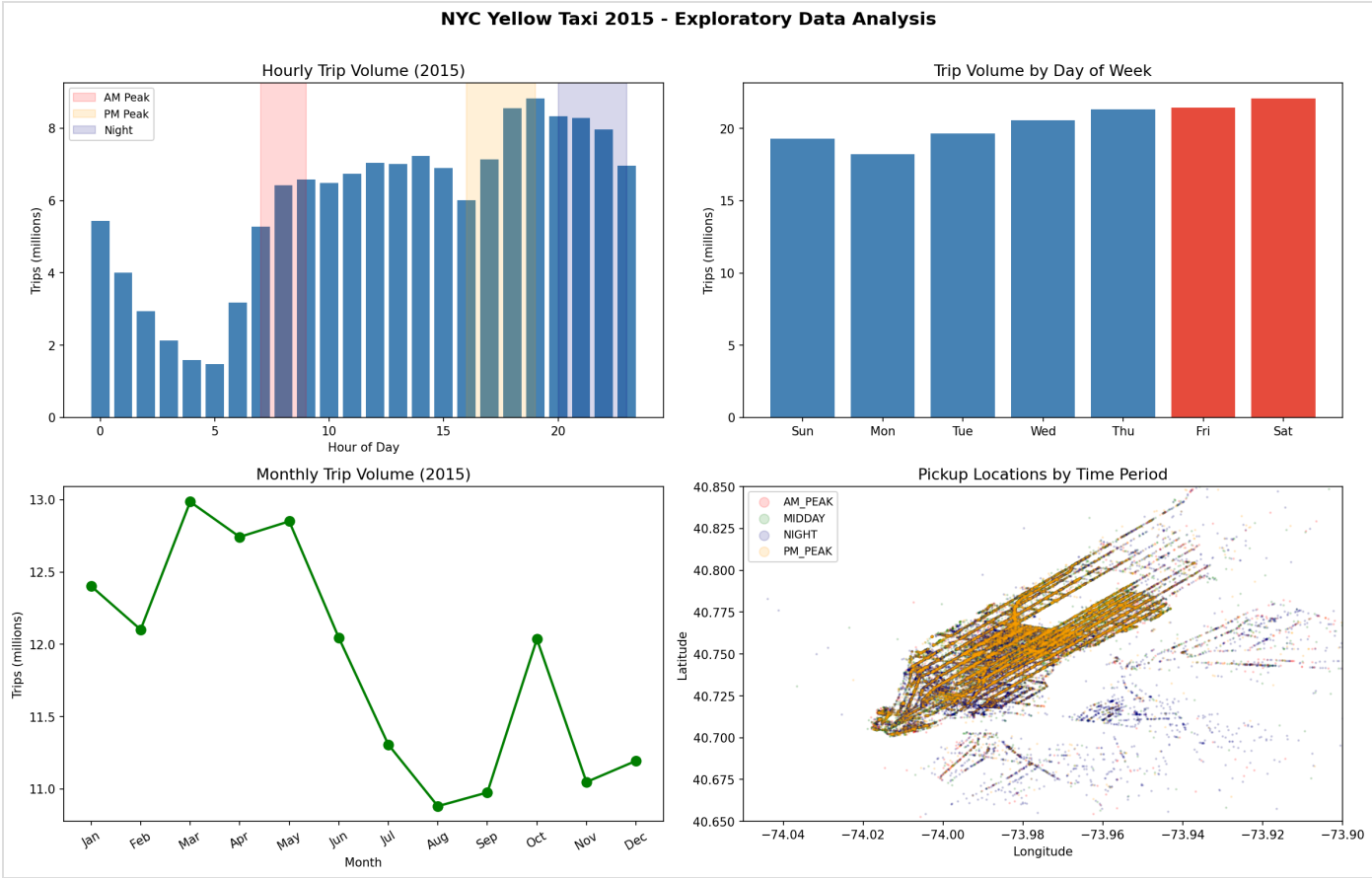


Fig. 0 — EDA overview: hourly/DOW/monthly volume + pickup scatter by period

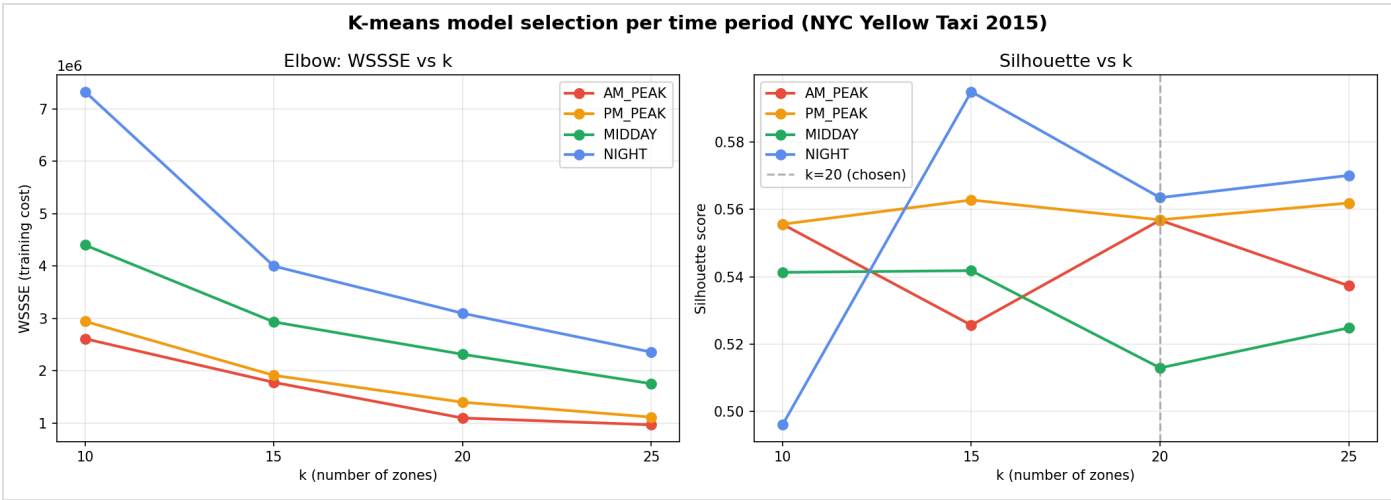


Fig. 1 — Elbow (WSSSE) + silhouette vs k, per period

Time-Adaptive Demand Zones (k=20 centroids per period)
NYC Yellow Taxi 2015 - zones shift only ~0.5-1 km between periods

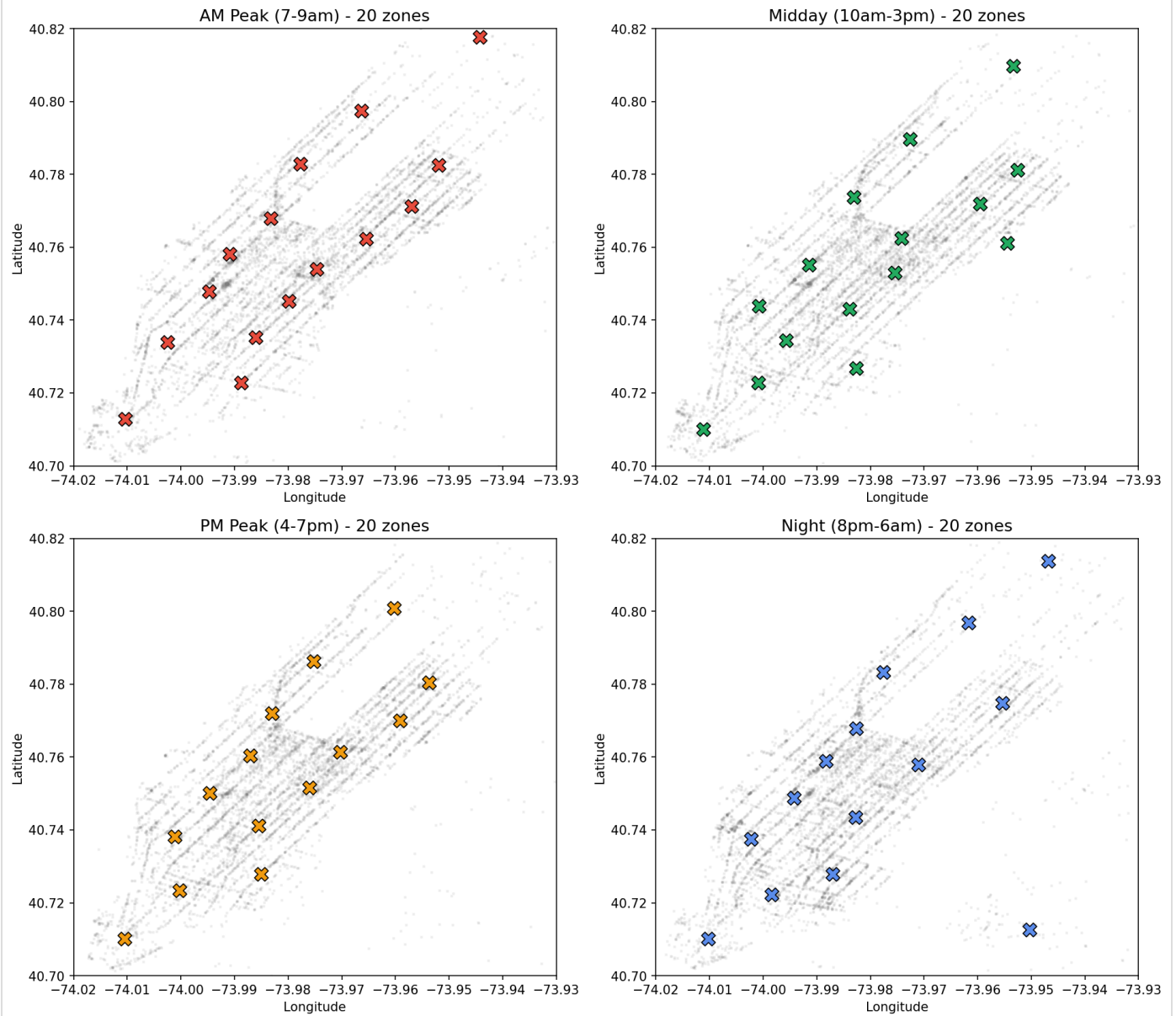


Fig. 2 — 20 demand zones per period over a pickup backdrop

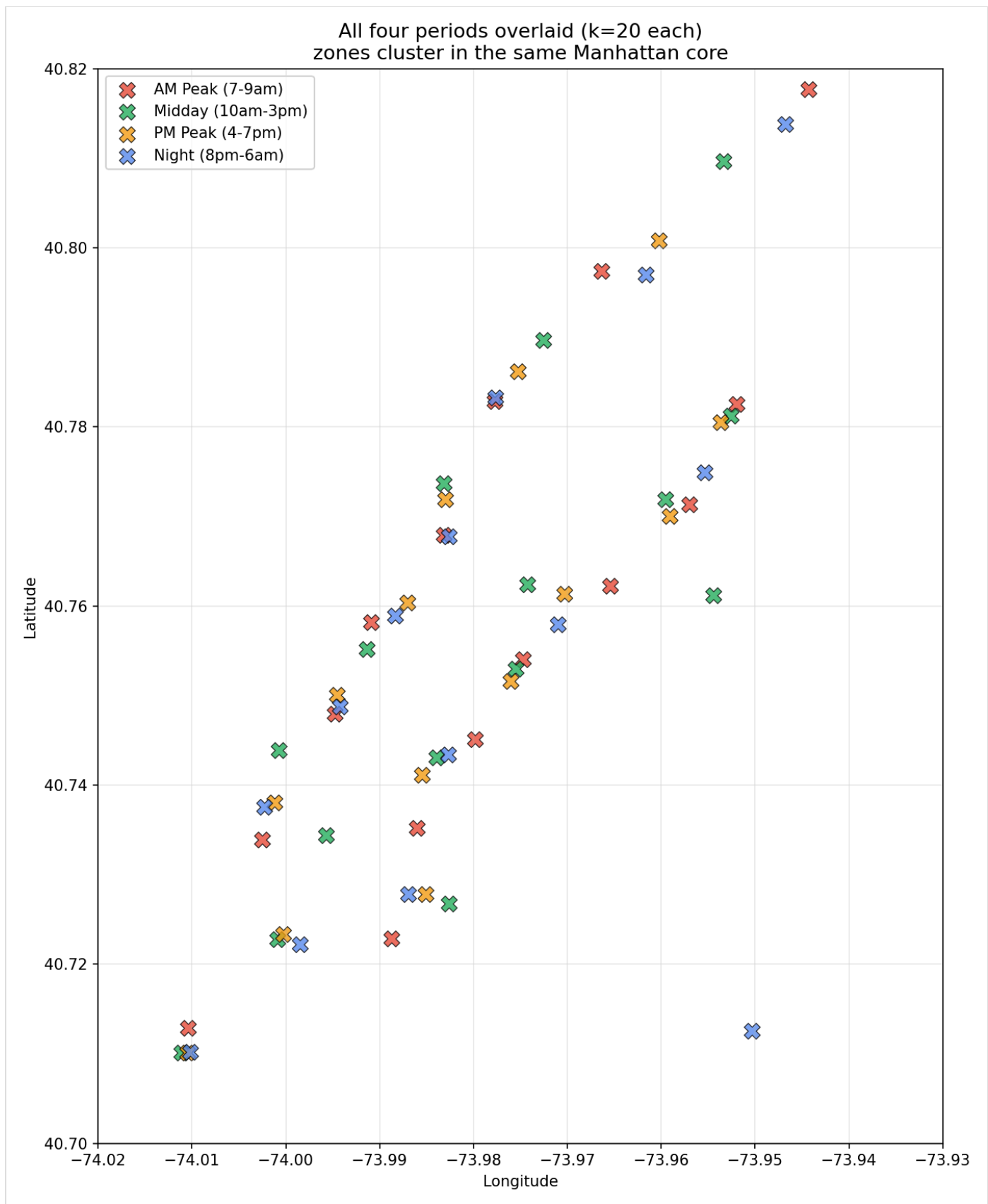


Fig. 2b — All four periods' centroids overlaid (stability)

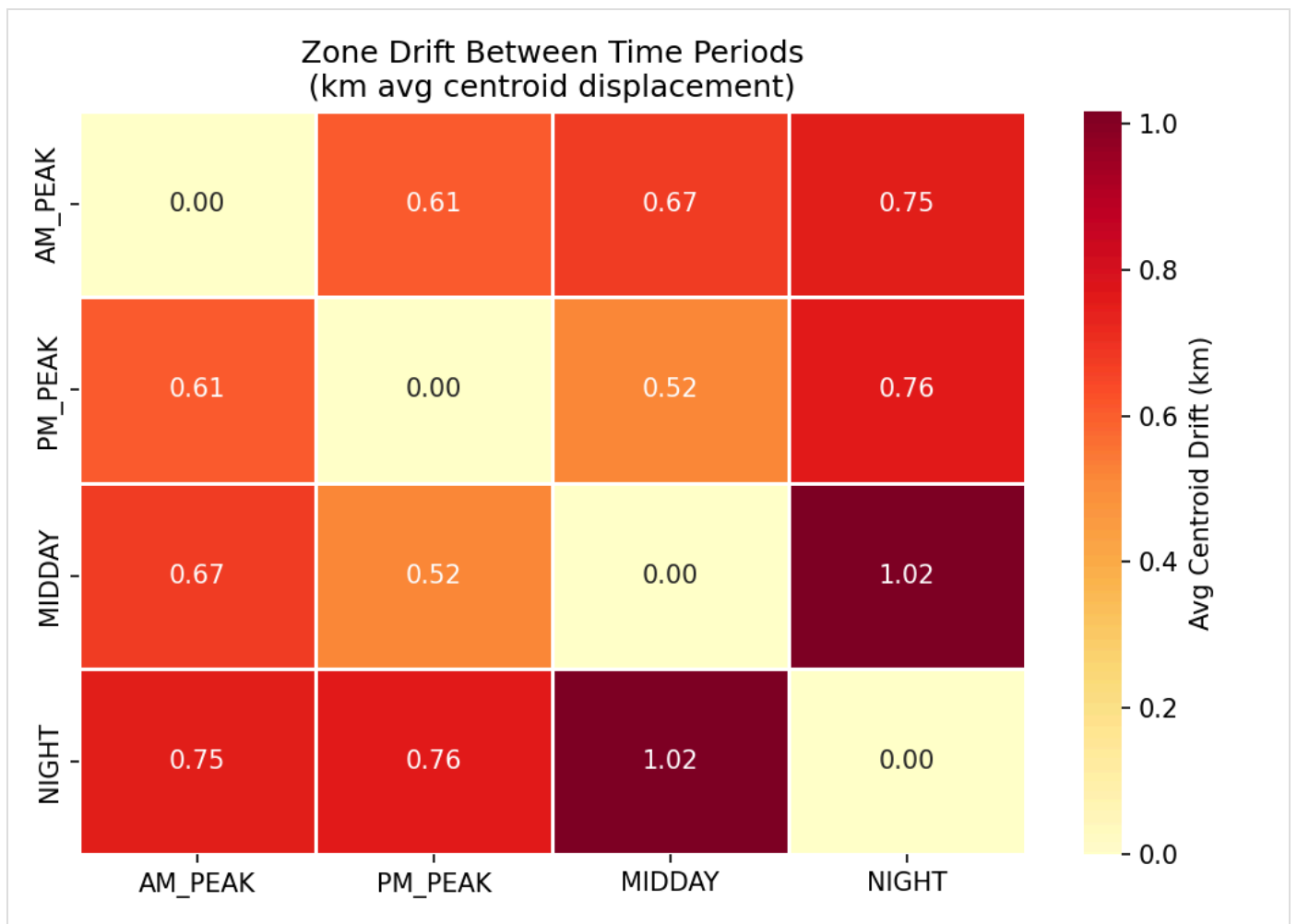


Fig. 3 — Zone drift heatmap: centroid displacement (km) between periods

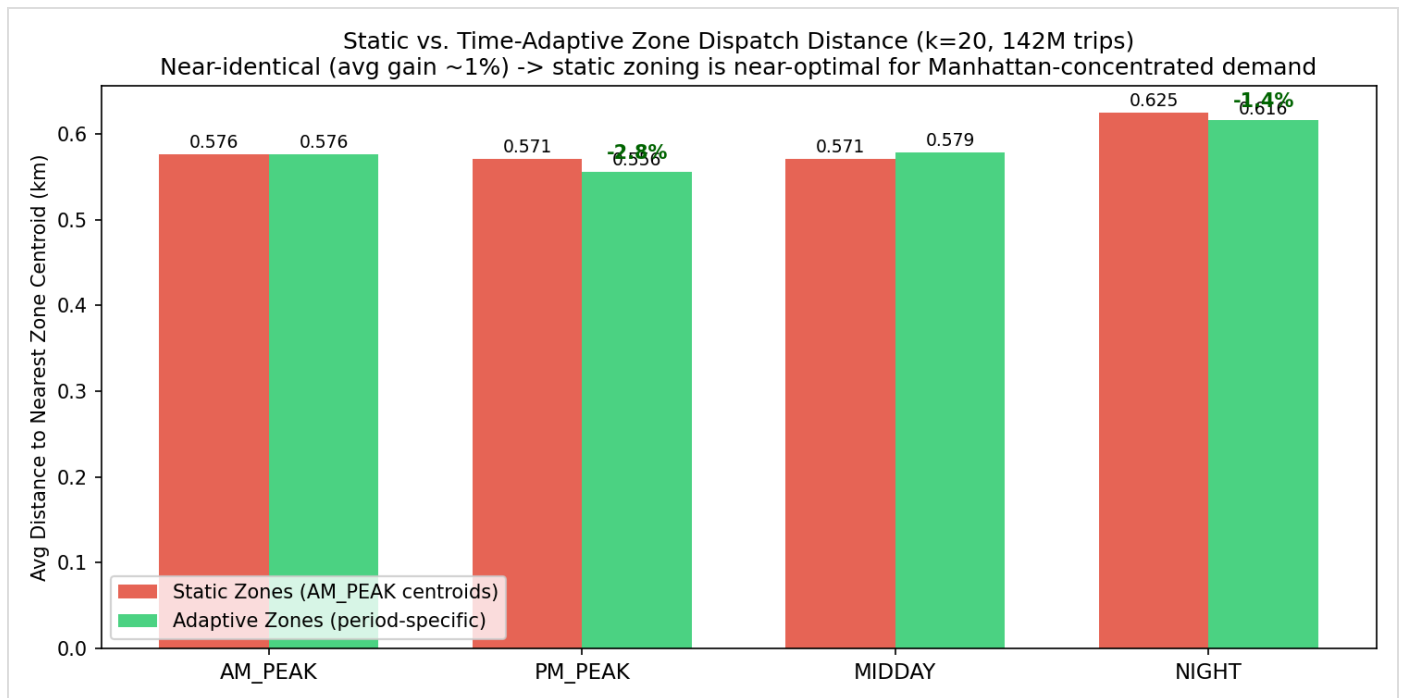


Fig. 4 — Routing: adaptive vs static dispatch distance (~1% gain)

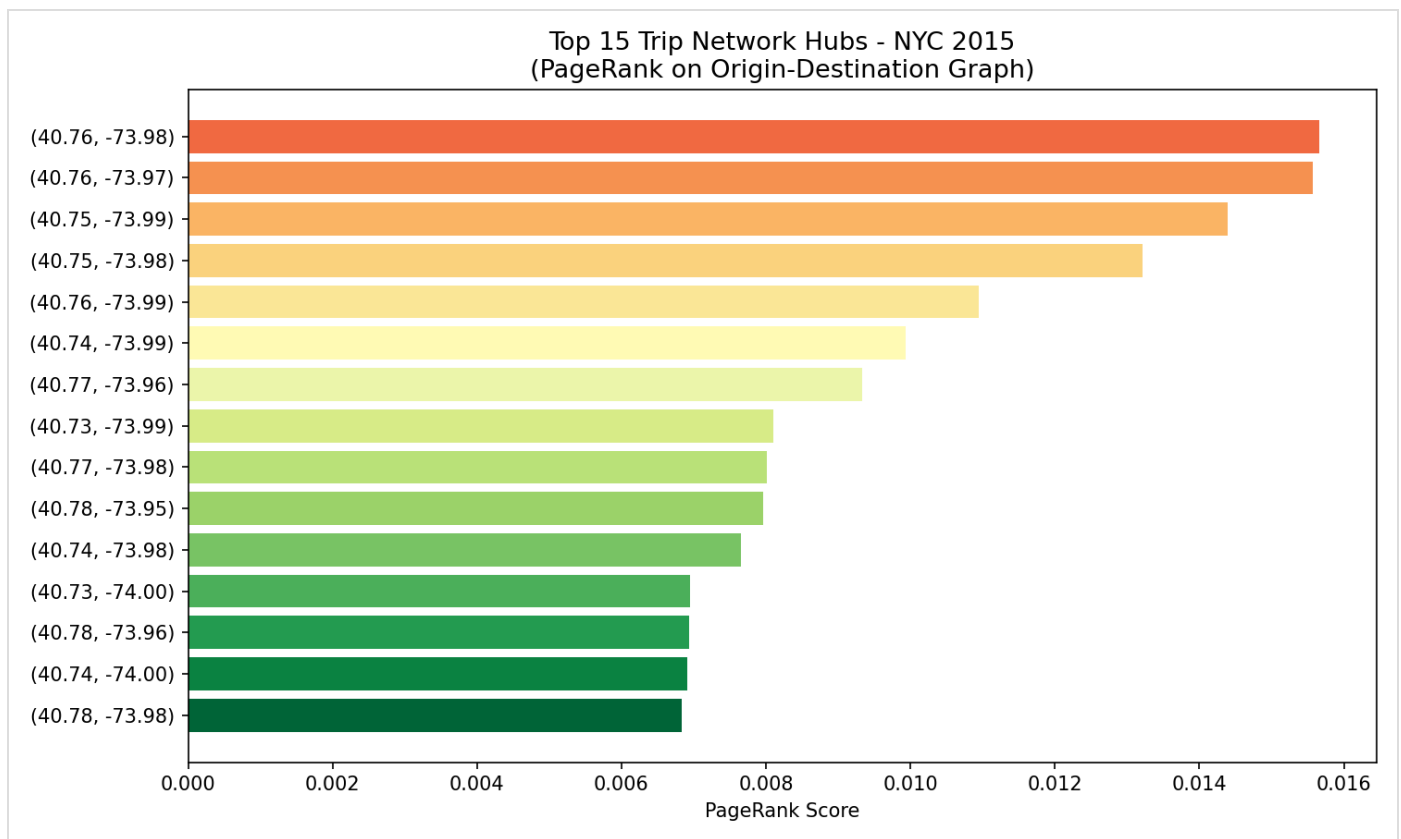


Fig. 5 — Top OD network hubs by PageRank (Midtown)

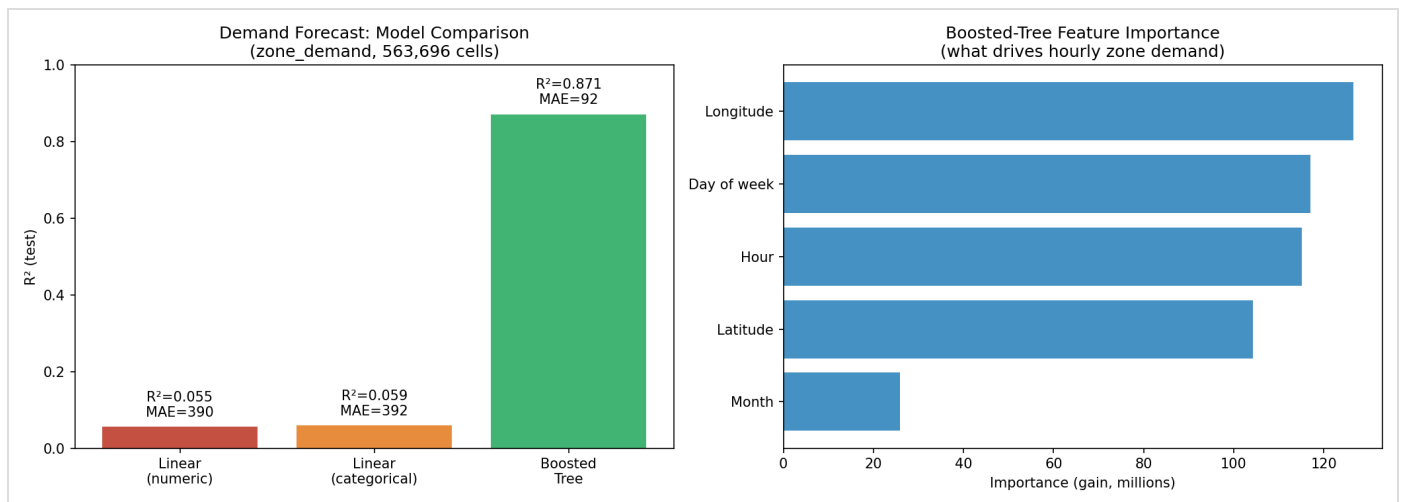


Fig. 6 — BQML model R^2 comparison + boosted-tree feature importance